Simplified Landing Dynamics Model

Progress report — steps 1–3: rigid-body dynamics, spring-damper contact, friction, and integrator stability
18 September 2026 · HeloSimWork / `SimpleLandingModel` · MATLAB R2026a · All numbers and figures from a single fresh run of `run_all_tests` (log: `report_run_log.txt`)

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1. Summary

3 / 3	**1.3e-13 m**	**2.4e-8**	**~21 mm/s**
verification suites passing (steps 1, 2, 3)	free-fall error vs. analytic (machine precision)	relative error, friction creep vs. analytic	physics-friction creep at dt = 0.01 (target: 4.4)

A minimal 6-DOF landing simulator was built from scratch in MATLAB as a controlled testbed for comparing **physics-based friction at dt = 0.001 s** against a future **machine-learning friction model at dt = 0.01 s**. Steps 1–3 of the build plan are complete and verified:

- **Rigid-body EOM and gravity** match the analytic free-fall solution to 10^{-13} m, including a tilted-body case that validates the rotation matrix and gravity transform together.
- **Spring-damper ground contact** settles to an independently computed static equilibrium to 6×10^{-6} N, conserves energy to 10^{-13} when undamped, and keeps roll and yaw *identically* zero under a symmetric drop.
- **Regularized Coulomb friction on a 5° incline** reproduces the analytic hold, creep, and slide-to-stop results to within 0.42% (slides) and 2×10^{-8} (steady creep).

> **Key finding.** The build plan's timestep analysis considered friction resisting *translation* only. Because the feet sit 1.8 m below the CG, friction also resists *roll*, and the roll inertia is small. The true stiffest friction mode is ~6× faster than planned (−3,454 rad/s at the original boundary layer), which made the planned dt = 0.001 physics reference **numerically unstable**. A laterally symmetric test cannot detect this — it passes perfectly — so the stability test seeds lateral motion deliberately.
>
> **Resolution:** the friction boundary layer was raised to match the UH-60 model's value (`LG_VBL` = 0.1 ft/s = 0.03048 m/s). The dt = 0.001 reference is now stable and verified. At dt = 0.01, physics friction remains unstable and chatters at ~21 mm/s of creep regardless of the requested boundary layer — the concrete gap an ML model at dt = 0.01 would need to close (target 4.4 mm/s).

2. Purpose and context

The parent project uses the PSUHeloSim UH-60 flight dynamics model. Its real-time simulator cannot run below dt ≈ 0.01 s without missing frame deadlines, but ground-contact and friction fidelity require a much smaller step. The research question is whether a learned friction model running at the coarse step can match physics friction running at the fine step.

The full UH-60 model is too coupled to answer that cleanly — a difference in output could come from the friction model or from any of a dozen interacting subsystems. This testbed strips the problem to a rigid body, three contact points, a planar floor, and friction, so that friction is the *only* source of numerical stiffness and any measured difference is attributable to it.

3. Model description

3.1 Geometry and parameters

A rectangular box stands in for the fuselage, with three legs in a helicopter-like tricycle stance: two forward legs splayed outboard and one aft. Legs are **rigid struts with a compliant foot** (penalty contact) — there is no unsprung mass. This is deliberate: it keeps the normal-direction dynamics soft (√(3k/m) ≈ 14 rad/s) so that friction regularization is the only stiff element.

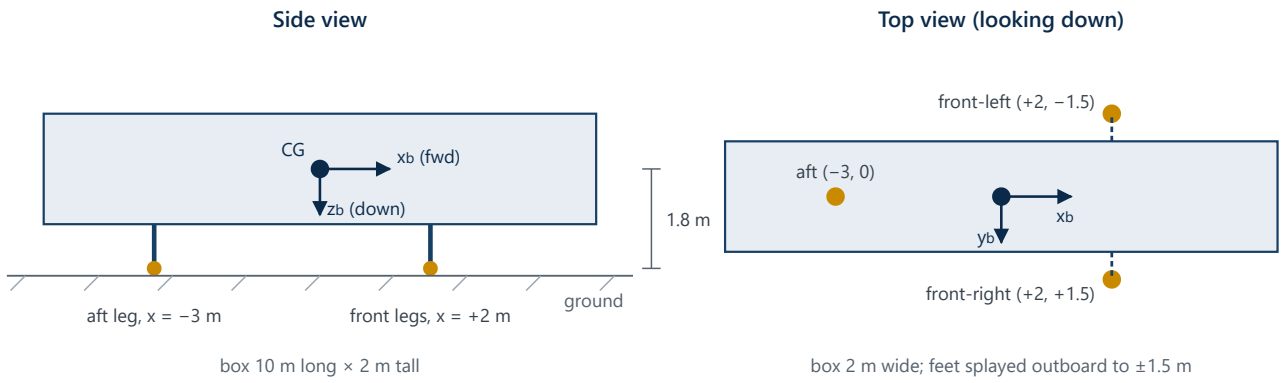


Figure 1. Model geometry (to scale, 30 px/m). Foot coordinates are in the body frame (x forward, y right, z down, origin at CG). Note $\Sigma x_i = +1$ m: the leg triangle is not centred under the CG, which produces unequal static leg loads and a small nose-up pitch (Section 6).

Parameter	Value	Basis
Mass m	1000 kg	Round number; not a scale model of the UH-60
Inertia I_{xx}, I_{yy}, I_{zz}	667, 8667, 8667 kg·m²	Solid box 10 × 2 × 2 m; I_{yy}/I_{xx} = 13 (rolls far more easily than it pitches)
Leg stiffness k (each)	6.54×10^4 N/m	Sized for 5 cm mean static deflection: k = mg / (3 × 0.05)
Leg damping c (each)	2800 N·s/m	Heave damping ratio ζ = 0.30 (visibly underdamped, errors easy to see)
Friction coefficient μ	0.6	Dry rubber-like; UH-60 model uses 0.5 (see Section 11)
Friction boundary layer v_{bl}	0.03048 m/s	**= 0.1 ft/s, matched to UH-60 `LG_VBL`** (was 0.01 m/s; Section 8)
Physics / ML timesteps	0.001 / 0.01 s	Study design
Integrator	fixed-step RK4	Hand-written: the study is *about* dt, so it must be explicit, not solver-chosen

3.2 Frames, state, and equations of motion

Inertial frame is NED (x North, y East, **z down**); altitude = −z. **Body frame** is x forward, y right, z down, origin at the CG; Euler sequence 3-2-1. SI units throughout. The state has 12 elements:

```
x = [ u v w | p q r | phi theta psi | xN yE zD ]     body velocity | body rates | Euler angles |
NED position

Vdot_b     = F_b / m  -  omega x V_b                          (Newton, rotating frame)
omegadot_b = I^-1 ( M_b  -  omega x (I omega) )              (Euler's equations)
phidot     = p + sin(phi) tan(theta) q + cos(phi) tan(theta) r
thetadot   =     cos(phi) q            - sin(phi) r
psidot     =    (sin(phi) q + cos(phi) r) / cos(theta)
Pdot_i     = R_b2i V_b
F_gravity_b = R_b2i' [0; 0; m g]      (derived from R, never written separately)
```

3.3 Contact and friction

The ground is a plane with unit normal **n** (pointing into the ground). For each leg, every derivative evaluation:

```
P_foot   = P_cg + R_b2i p_foot_b                    foot position, inertial
delta    = n . (P_foot - p0)                         penetration along the terrain normal (> 0 =
contact)
V_foot   = R_b2i (V_b + omega x p_foot_b)
N        = max( k delta + c (n . V_foot), 0 )        ground pushes, never pulls
v_t      = V_foot - (n . V_foot) n                   tangential slip velocity
F_t      = -mu N v_t / max(|v_t|, v_bl)              regularized Coulomb friction  (swappable)
F_foot_b = R_b2i' ( F_t - N n )                      ONE inertial vector, rotated to body BEFORE
summing
M_foot_b = p_foot_b x F_foot_b                       moment about the CONTACT POINT
```

The last two lines carry forward the lesson from the UH-60 `LandingGear.m` bug found earlier in this project, where a terrain-frame friction force was concatenated with a body-frame strut force and the friction moment was taken about the axle rather than the contact patch. The friction model is passed in as a function handle, so the ML comparison is a one-line change with every other part of the model held bit-for-bit identical.

4. Verification approach

The guiding rule is **use an exact pass criterion wherever exactness is physically achievable**, because a tight tolerance surfaces mistakes that a loose one hides. Three examples: RK4 integrates constant acceleration exactly, so free fall is checked at machine precision; the leg geometry is exactly laterally symmetric, so roll and yaw under a symmetric drop must be *identically* zero; and energy conservation is checked separately in a smooth regime and across contact discontinuities so the two error sources can be told apart. Every run also passes a common set of safety checks: no NaN/Inf, no adhesion ($N \geq 0$), contact flags consistent with force, and friction never exceeding μN.

5. Step 1 — free fall (contact disabled)

Pass criterion: the inertial trajectory must match $P(t) = P(0) + V(0)t + \frac{1}{2}[0, 0, g]t^2$ to machine precision.

Case	Max position error	Attitude drift	Body rates	Result
1a — level, at rest	1.33×10^{-13} m	0	0	PASS
1b — level, translating	1.33×10^{-13} m	0	0	PASS
1c — tilted (20°, −15°, 30°), translating	2.78×10^{-13} m	0	0	PASS

Case 1c is the meaningful one: with gravity resolved through a rotated body frame and no moments acting, the inertial acceleration must still be exactly [0, 0, g] and the attitude must stay frozen. That validates the rotation matrix and the gravity transform simultaneously.

6. Step 2 — spring-damper contact (friction off)

The static equilibrium is not level. With feet at x = +2, +2, −3 m, equal leg loads would leave a 3270 N·m pitch moment, so the aft leg carries more load and the body sits slightly nose-up. The exact equilibrium was solved independently of the model code (Newton iteration on hand-derived force and moment balance) and the simulation compared against it:

Check	Result	Tolerance	
Leg loads at equilibrium (FL, FR, aft)	2937.62, 2937.62, 3934.77 N	—	
Penetrations (FL, FR, aft); mean	44.92, 44.92, 60.16 mm; 50.00 mm	—	
Equilibrium pitch	0.003049 rad (0.175°)	—	
2A — settled state vs. exact equilibrium	ΔN 6.2×10^{-6} N, $\Delta\theta$ 1.8×10^{-11} rad	10^{-2}, 10^{-7}	PASS
2B — force balance $\lvert\Sigma N - mg\rvert$	1.9×10^{-6} N	10^{-3}	PASS
2B — moment balance $\lvert M\rvert$	1.8×10^{-5} N·m	10^{-2}	PASS
2C — hold at equilibrium for 2 s	2.3×10^{-11} m, 2.6×10^{-11} rad	10^{-4}	PASS
2D — roll and yaw, symmetric drop	identically 0	10^{-10}	PASS
2E — energy drift, c = 0, continuous contact	1.9×10^{-13} (relative)	10^{-8}	PASS
2F — energy drift, c = 0, repeated bounce	2.2×10^{-6} (relative)	10^{-3}	PASS

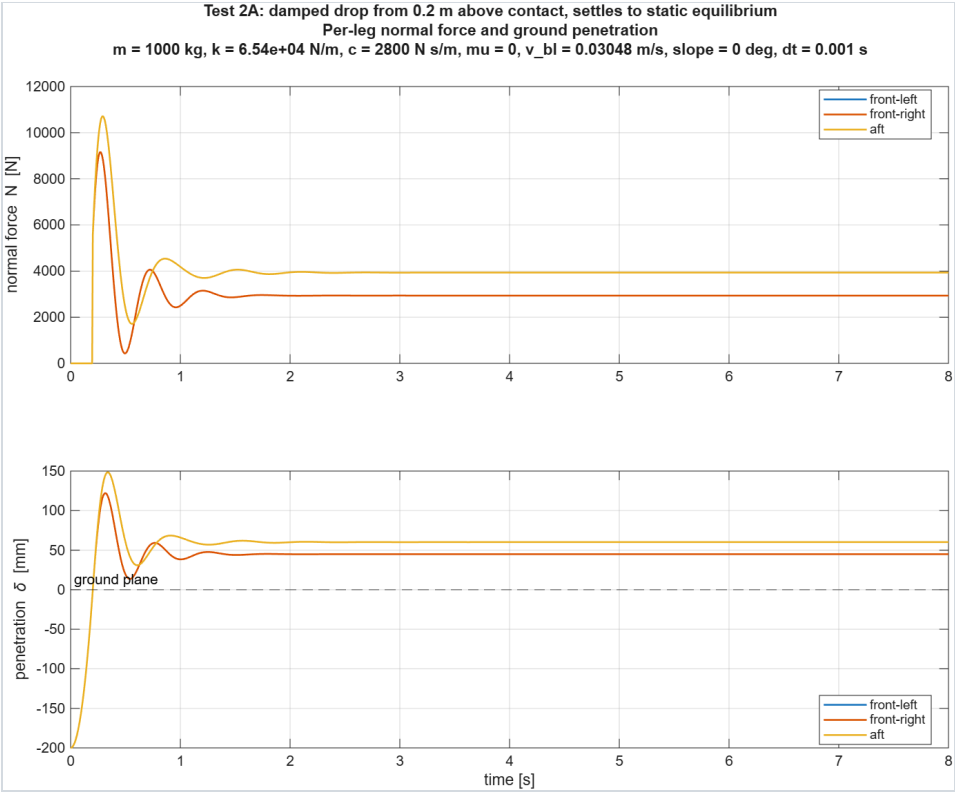


Figure 2. Test 2A — damped drop from 0.2 m above first contact. Force is zero until touchdown at ~0.2 s; the aft leg peaks and settles higher than the front legs; the response settles by ~2 s. The front-left trace is completely hidden beneath front-right — visual confirmation that the lateral symmetry holds exactly.

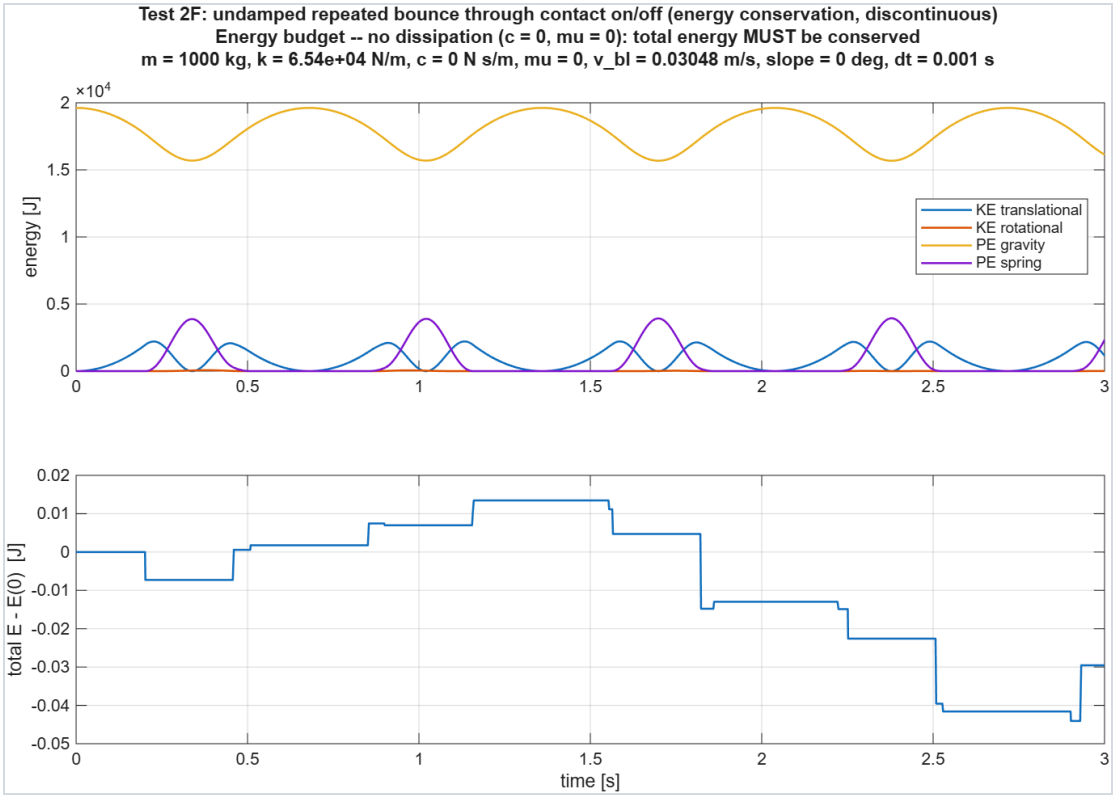


Figure 3. Test 2F — undamped repeated bounce. Top: energy exchanges between gravity, kinetic and spring potential with no decay. Bottom: total-energy error changes only in *steps at the contact on/off instants* and is flat in between (±0.04 J out of 19,620 J). This localizes the 2.2×10^{-6} drift to fixed-step integration across the contact discontinuity rather than to the integrator generally — consistent with the 1.9×10^{-13} drift in the smooth continuous-contact case (2E).

7. Step 3 — friction on a 5° incline

The contact model was generalized from a flat z = 0 floor to an arbitrary plane. On flat ground the generalized model reproduces every step-1 and step-2 result **bit-identically**. The incline rises toward North; μ = 0.6, v_{bl} = 0.03048 m/s. Validation runs use dt = 0.001, but only after a linearization gate confirms that step is stable for the configured boundary layer (max |R(hλ)| = 0.998 < 1).

> **Why regularized friction creeps.** To generate the downhill-resisting force mg sin α inside the boundary layer, the law F = μN·v/v_{bl} needs a non-zero sliding speed. In steady state this gives an exact prediction: **$v_{creep} = v_{bl} \tan\alpha / \mu = 4.444$ mm/s**. The vehicle cannot hold perfectly still — this is the regularization artifact the whole study is about.

7.1 Test 3A — settle and hold

Check	Simulated	Analytic	Error	
Creep speed (downhill)	4.444424×10^{-3} m/s	4.444424×10^{-3} m/s	2.4×10^{-8} rel.	PASS
Sum of normal forces	9772.6700 N	mg cos α = 9772.6700 N	4.4×10^{-7} N	PASS
Total friction, uphill	854.9978 N	mg sin α = 854.9978 N	1.4×10^{-6} N	PASS
Force / moment balance	1.4×10^{-6} N / 6.3×10^{-6} N·m	0	tol 10^{-2}	PASS
Holding, not sliding: max $\lvert F_t\rvert$ / μN	0.146	< 1	—	PASS
Roll, yaw, v, p, r	identically 0	0	tol 10^{-10}	PASS

The body settles at 5.256° pitch — the 5° slope plus a 0.256° offset from the uneven leg loads, which the uphill friction acting below the CG increases relative to flat ground (leg loads 2769.97 / 2769.97 / 4232.73 N).

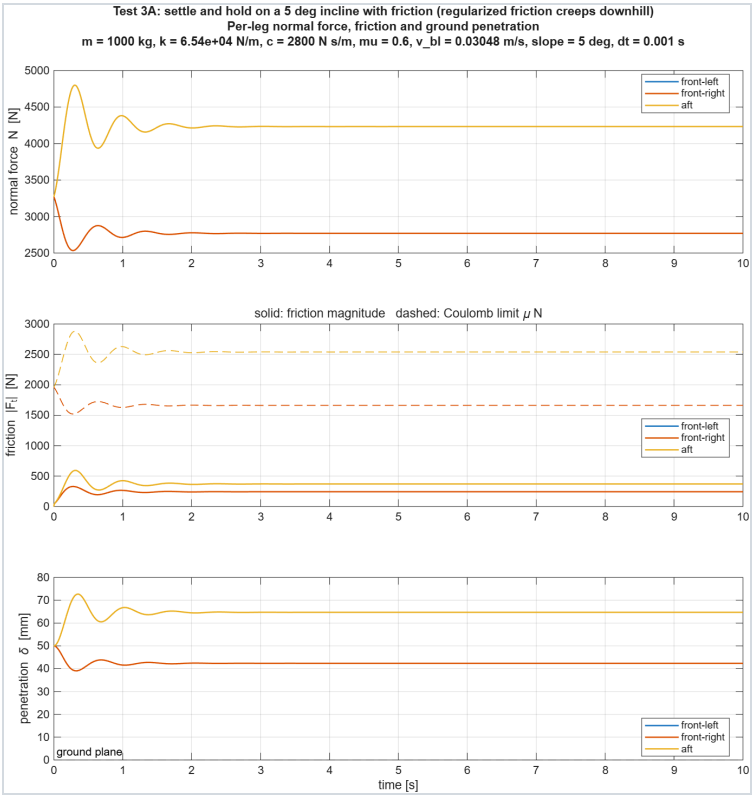


Figure 4. Test 3A — settle and hold on the 5° incline. Middle panel: each leg's friction (solid) sits well below its Coulomb limit μN (dashed) at a ratio of 0.146 = v_{creep}/v_{bl}, the signature of holding inside the boundary layer rather than sliding.

7.2 Test 3B — slide to a stop from 3 m/s

Distance is measured until speed first drops below 0.10 m/s — more than 3× v_{bl}, so the boundary layer plays no part. The 2% tolerance reflects the quasi-static assumption ΣN = mg cos α, which the pitch/heave transient from sudden friction onset violates slightly.

Case	Deceleration	Distance: sim / analytic	Error	Time: sim / analytic	
Downhill (tail-first)	$g(\mu \cos\alpha - \sin\alpha) = 5.009$ m/s^2	0.8984 / 0.8975 m	0.10%	0.578 / 0.579 s	PASS
Uphill (nose-first)	$g(\mu \cos\alpha + \sin\alpha) = 6.719$ m/s^2	0.6662 / 0.6690 m	0.42%	0.435 / 0.432 s	PASS

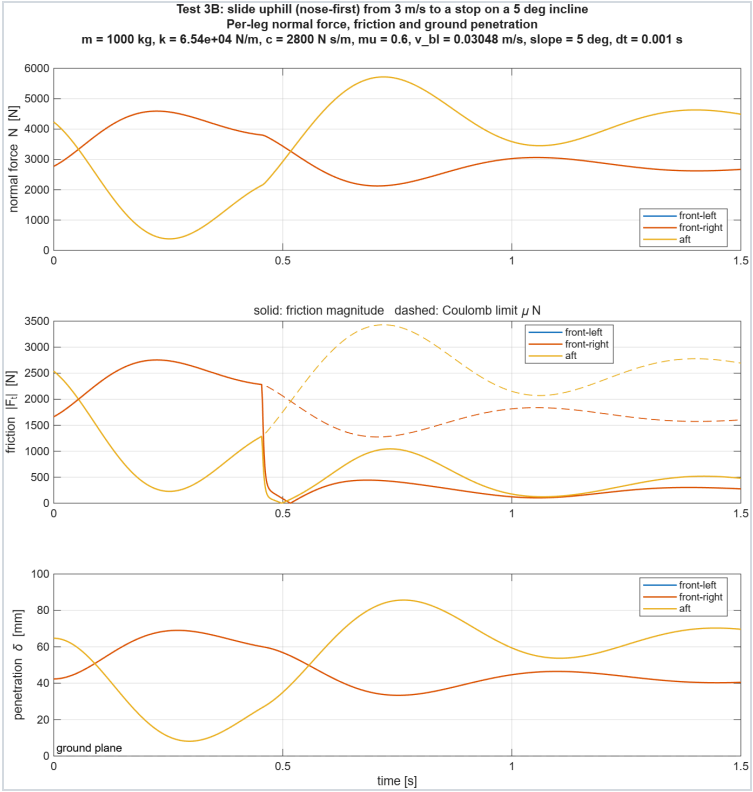


Figure 5. Test 3B — nose-first uphill slide. While sliding (0–0.44 s) each friction trace lies *on* its μN limit (fully sliding); at the stop it drops below the limit into the holding regime. Braking friction below the CG pitches the nose down and transfers load onto the front legs; the aft leg unloads to 380 N at its lightest but never leaves the ground.

8. Integrator stability finding

8.1 What the plan assumed, and what is actually true

The build plan sized friction stiffness from translation alone: $c_{eff} = \mu N / v_{bl}$ acting on the full 1000 kg, giving −589 rad/s at $v_{bl} = 0.01$ and the conclusion that dt = 0.001 is stable. That is incomplete. Friction acts at the feet, 1.8 m below the CG, so it also drives rotation. The effective mass seen by lateral friction combines translation and roll:

```
1/m_eff = 1/m + z^2/Ixx = 1/1000 + 1.8^2/667   ->   m_eff = 171 kg      (roll coupling factor 5.86)
```

Linearizing the full 12-state model at the slope equilibrium (numerical Jacobian) gives a fastest pole of **−3,454 rad/s** at $v_{bl} = 0.01$ (lateral/roll), plus −807 rad/s (longitudinal/pitch). For RK4 the step is stable only

if $|h\lambda| < 2.785$ on the real axis, so dt = 0.001 ($|h\lambda| = 3.45$) is unstable.

8.2 Why a symmetric test cannot see it

On a slope that rises along the body's x-axis, the geometry is exactly laterally symmetric: v, p, r, roll and yaw stay at *exactly* zero in floating point. The unstable lateral mode is never excited, so an unseeded run at the unstable setting shows zero lateral motion and looks correct. Test 3C therefore starts every run with a 10^{-4} m/s lateral velocity seed. The classification criterion is physical: a stable integration of a dissipative system must shed a lateral disturbance; an unstable one amplifies it. The instability does not diverge to NaN — once foot slip exceeds v_{bl}, friction saturates, stops being stiff, and the model settles into bounded chatter.

8.3 Test 3C — prediction vs. seeded simulation

Configuration	v_{bl}	dt	$\lvert\lambda\rvert_{max}$	max|R(hλ)|	Predicted	Seed growth	Creep: sim / analytic	Match
Physics reference (UH-60 v_{bl})	0.0305	0.001	1143	0.998	stable	0.06×	4.444 / 4.444 mm/s	PASS
Control run B (UH-60 v_{bl})	0.0305	0.01	1143	518	unstable	298×	21.24 / 4.44 mm/s	PASS
Original tight v_{bl}	0.01	0.001	3454	2.57	unstable	8.1×	—*	PASS
Original tight v_{bl}, halved dt	0.01	0.0005	3454	0.999	stable	0.07×	—*	PASS
Loosened until stable at dt = 0.01	0.15	0.01	244	0.978	stable	0.03×	21.91 / 21.87 mm/s	PASS

"Match" means the observed behaviour (seed grew or decayed) agrees with the linear prediction. *The v_{bl} = 0.01 runs start from the v_{bl} = 0.0305 equilibrium and are still converging after 2 s, so their creep values are not quoted as results.

Stability boundary. Bisection on the linear prediction gives the smallest stable boundary layer for each step, and the smallest creep physics friction can therefore deliver:

Timestep	Smallest stable v_{bl}	Minimum achievable creep on 5°
dt = 0.001	0.0124 m/s	1.81 mm/s
dt = 0.01	0.1303 m/s	18.99 mm/s

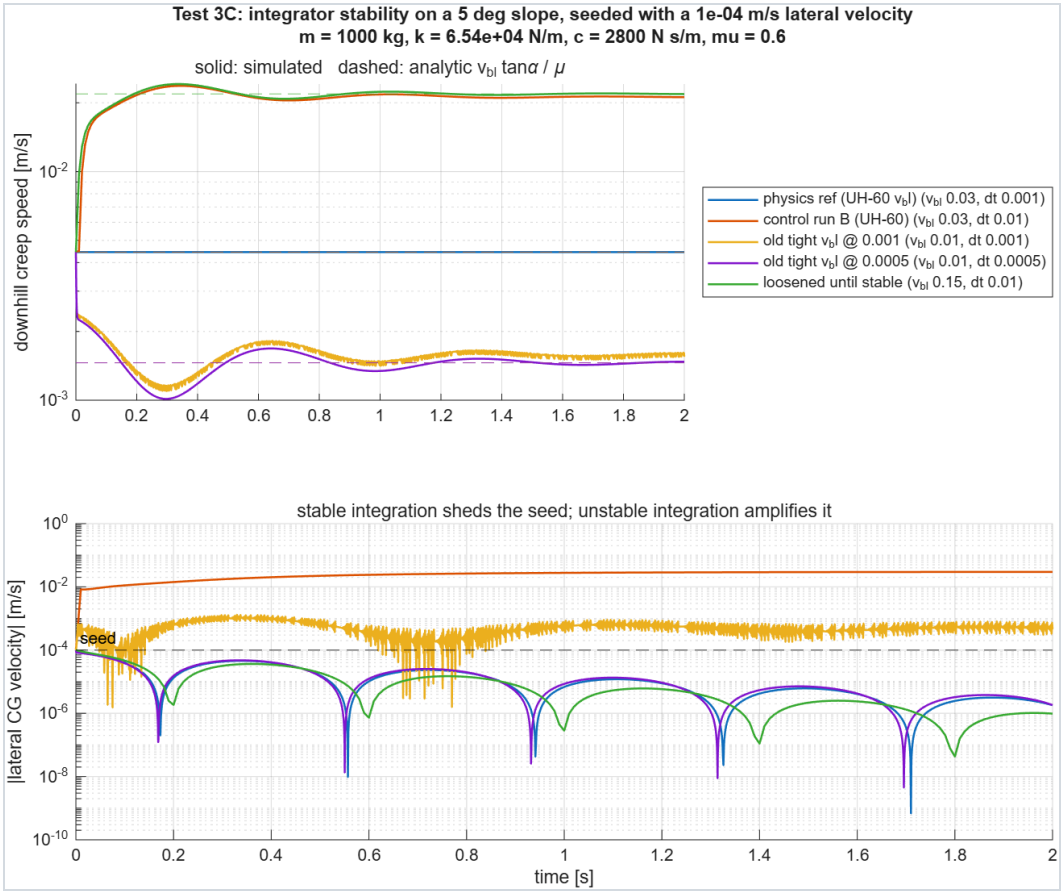


Figure 6. Test 3C — seeded stability runs. Top: at dt = 0.01 every physics-friction run converges to ~21 mm/s of creep; only the genuinely stable v_{bl} = 0.15 run lands on its own analytic line, while the UH-60 v_{bl} run sits ~5× above its target of 4.44 mm/s. Bottom: stable integrations damp the 10^{-4} m/s seed below the dashed line; unstable integrations amplify it into sustained chatter.

> **Implication for the ML study.** Below its stability limit, physics friction at dt = 0.01 does not blow up — it chatters at roughly the creep of the stability limit, *whatever boundary layer is requested*. So at the real-time step, physics friction cannot hold the vehicle tighter than ~19–21 mm/s of creep on a 5° slope, against 4.44 mm/s from the dt = 0.001 reference. That factor of ~5 is the measurable gap a learned friction model at dt = 0.01 must close.

9. Comparison with the UH-60 model

The UH-60 model runs its landing gear at dt = 0.001 without this friction instability. Applying the same effective-stiffness estimate $\lambda \approx (\mu \Sigma N/m)/v_{bl} \times (1 + m\, z^2/I_{xx})$ to both models:

Factor	This model (original)	UH-60 model	UH-60 softer by
Boundary layer v_{bl}	0.01 m/s	0.1 ft/s = 0.0305 m/s	3.05×
Friction coefficient μ	0.6	0.5	1.2×
ΣN/m, full weight on gear	9.8 m/s²	10.5 m/s²	0.93×
Roll coupling $1 + m z^2/I_{xx}$	5.86	≈ 4.48	1.31×
Fastest friction pole	**−3,454 rad/s** (linearized)	**≈ −770 rad/s** (estimate)	**~4.5×**

At dt = 0.001 the UH-60 value gives |hλ| ≈ 0.77, well inside its `ode3` solver's limit (~2.51). The dominant difference is the boundary layer; the UH-60 source comment on `LG_VBL` reads "0.1 ft/sec appears to eliminate chattering". Matching that value is what stabilized this model's reference. Caveats: the UH-60 figure is a hand estimate rather than a linearization, the UH-60 has not been tested with a lateral seed, and during landings the rotor carries much of the weight, which lowers the friction pole further.

10. Corrections made along the way

Several of the build plan's assumptions, and one of my own cross-checks, were wrong. Each was caught by a tight tolerance or a direct check and is recorded here and in the repository.

What was assumed	What is true	How it was caught
Step 2 drop "settles level at 5 cm"	Settles 0.175° nose-up with unequal loads; only roll and yaw are exactly level; mean deflection is 5.000 cm	Leg geometry: $\Sigma x_i = +1$ m
First hand-derived equilibrium ($\Sigma x_i N_i = 0$)	Off by ~11 N — it dropped the $z_i \sin\theta$ moment-arm term (53.8 N·m); the *simulation* was right	Tight 2A tolerance; replaced by an exact Newton solve
Friction pole −589 rad/s; dt = 0.001 stable at $v_{bl} = 0.01$	−3,454 rad/s via roll coupling; unstable at dt = 0.001	Linearization of the full model
$v_{bl} = 0.05$ stable at dt = 0.01	Unstable; needs $v_{bl} \geq 0.130$	Linearization + seeded runs
A clean symmetric run means the step is stable	Symmetry never excites the unstable mode	Seeded vs. unseeded comparison
Energy rise or friction sign reversal flags instability	Neither does here (downhill creep keeps releasing energy); seed growth does	Diagnostic runs before choosing the criterion

11. Next steps

1. **Run the three-case ML experiment** as designed: (A) physics, dt = 0.001, reference — now verified stable; (B) physics, dt = 0.01, control — verified to chatter at ~21 mm/s; (C) ML, dt = 0.01 — target the reference's 4.44 mm/s creep and slide-to-stop distances. Seed lateral motion in every stability-relevant run.
2. **Train on windowed friction, not instantaneous force.** The ML model should predict the effective friction over each 0.01 s window given the state at its start. Learning instantaneous force from

instantaneous velocity would reproduce the same stiff dynamics and gain nothing at the coarse step.

3. **Decide on μ.** This model uses 0.6; the UH-60 uses 0.5. Matching it would soften the fastest pole to roughly −950 rad/s and raise creep on 5° to ~5.3 mm/s.
4. **Check the UH-60 model the same way.** Its friction stability at dt = 0.001 is estimated, not measured, and it has never been run with a lateral seed.
5. **Scale-up** (after the ML result): per-leg compression DOF with unsprung mass (reintroduces genuine normal-direction stiffness), nonlinear gas-spring leg law, richer friction (LuGre/brush), rotor thrust input.

Appendix — files and how to run

Run everything: `matlab -batch "run_all_tests"` from the `SimpleLandingModel` folder. Runs all three suites and writes 25 figures to `figures/`.

File	Role
`rot_b2i.m`	Body→inertial rotation — the single definition, called everywhere
`landing_params.m`	All parameters in one struct (v_{bl} = 0.1 ft/s)
`make_terrain.m`	Planar terrain from slope and uphill heading → normal vector
`landing_eom.m`	12-state equations of motion, xdot = f(F, M)
`contact_forces.m`	Per-leg contact relative to the terrain plane → body force and moment
`friction_physics.m`	Regularized Coulomb friction — the swappable interface
`rk4_step.m, run_landing_sim.m`	Fixed-step RK4 and the logging driver
`linearize_eom.m`	Numerical Jacobian and eigenvalues, for stability prediction
`landing_energy.m, check_sanity.m`	Energy bookkeeping and the common safety checks
`test_step1_freefall.m, test_step2_contact.m, test_step3_friction_slope.m`	Verification suites
`run_all_tests.m, plot_landing_results.m`	Runner and figure generation
`PLAN.md, README.md`	Design plan (with the Section 6 correction) and current status